CYBERSECURITY CAPSTONE PORTFOLIO

Phishing Investigation, Threat Detection and Incident Response Using Wazuh

PART A — Phishing Investigation | Prepared by: Alex Osale

PART B — Wazuh Threat Detection | Prepared by: Alex Osale and Raheemah Lartey

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Tools: Wazuh • Ubuntu • Kali Linux • YARA 4.5.5 • Apache • Rootcheck • VirtualBox • pfSense

Portfolio purpose: Present evidence-based SOC analytical work, the detection workflow, results, verification, and limitations of controlled laboratory exercises.

Executive summary

This portfolio combines two related cybersecurity investigations. Part A examines four instructor-provided synthetic phishing emails using static header, sender, URL, attachment and social-engineering analysis. Part B documents practical monitoring and threat-detection exercises using an Ubuntu endpoint, Wazuh, YARA, Rootcheck and Apache access logs. Together, the exercises demonstrate alert triage, indicator examination, evidence documentation, careful incident classification and response planning.

Evidence standard: Findings are limited to the supplied samples and laboratory observations. Reported email authentication results were not independently verified; no credential theft, payment loss, production compromise, automatic YARA remediation or successful Shellshock exploitation was established.

Project structure

Workstream	Scope	Contributors
Part A	Four synthetic phishing scenarios; email analysis and response recommendations	Alex Osale
Part B	Tasks 5–8: Wazuh monitoring, YARA, Rootcheck and Shellshock-style request	Alex Osale and Raheemah Lartey

SOC workflow demonstrated

Preparation → Evidence collection → Detection → Triage → Analysis → Disposition → Response → Verification → Reporting

PART A — PHISHING EMAIL INVESTIGATION

Prepared by: Alex Osale

Method and limitations

Reviewed supplied sender and envelope identities, Reply-To fields, stated transport IPs, reported SPF/DKIM/DMARC results, deceptive links, attachment types, urgency cues and potential business impact. The four samples are synthetic; their header claims are not independently authenticated and no live domain or reputation verification was performed.

PA-01 — Microsoft account suspension

Assessment: High-confidence suspected credential phishing

Sender/routing evidence: From account-security@microsoft.com; Return-Path noreply@mailer-srv-dispatch.xyz; Reply-To ms-support@account-ms365-verify.top; stated sender IP 185.220.101.47.

Key indicators: Destination login-microsoft365-verify[.]top and deceptive support hyperlink; 24-hour deletion threat; account_verification.pdf; reported SPF/DKIM/DMARC failures.

Recommended action: Preserve email and PDF; avoid embedded links; verify independently through official Microsoft account access; review identity and email logs if interaction is suspected.

PA-02 — Supplier invoice and payment diversion

Assessment: High-confidence suspected BEC-style payment diversion

Sender/routing evidence: From ap.billing@deltasupplies-corp.com; Return-Path billing@fast-invoice-send.ru; Reply-To payments@fast-invoice-send.ru; stated IP 194.165.16.101.

Key indicators: Invoice DS-2026-04418 requests USD 47,250 and new wire details; Latvian IBAN LV82HABAL0000034567891; attached PNG; authentication failures reported, not verified.

Recommended action: Hold payment; validate supplier details using previously established contacts; involve Finance and SOC; initiate recall if payment occurred.

PA-03 — DHL failed delivery

Assessment: High-confidence suspected delivery-fee phishing

Sender/routing evidence: From noreply@dhl.com; Return-Path tracking@dhl-delivery-info.xyz; stated IP 176.111.174.55.

Key indicators: Nonofficial dhl-delivery-info[.]xyz redelivery link; USD 2.99 fee; tracking 1928374650; ShippingDocs_1928374650.zip. Archive contents and maliciousness not established.

Recommended action: Do not pay or open archive on ordinary workstation; verify parcel through official courier channel; preserve original artefacts.

PA-04 — Bank of America security alert

Assessment: High-confidence suspected banking credential phishing

Sender/routing evidence: From security.alert@bankofamerica.com; Return-Path noreply@secure-boa-alerts.cc; Reply-To security-noreply@secure-boa-alerts.cc; stated IP 91.108.56.122.

Key indicators: Password-reset link at secure-boa-alerts[.]cc; 12-hour urgency; alleged account access from 185.220.101.47; attached PNG; authentication failures reported, not independently validated.

Recommended action: Use independently accessed banking app or official website; preserve message; assess interaction and protect account if credentials were entered.

Cross-case correlation and caution

The synthetic samples reuse 185.220.101.47 (PA-01 stated sender IP; PA-04 alleged login source) and 91.108.56.122 (PA-01 alleged sign-in source; PA-04 stated sender IP). This pattern is a lead within the dataset, not proof of shared infrastructure, real-world malicious reputation or common attacker attribution.

PART B — WAZUH THREAT DETECTION AND INVESTIGATION

Prepared by: Alex Osale and Raheemah Lartey

Lab architecture

System	Role / details
Wazuh manager	Wazuh 4.14.6; reported manager address 192.168.1.11
Ubuntu endpoint	Ubuntu 24.04.3 LTS; Wazuh agent 001; 192.168.1.12
Kali Linux	Authorised attack-simulation source; 192.168.1.14

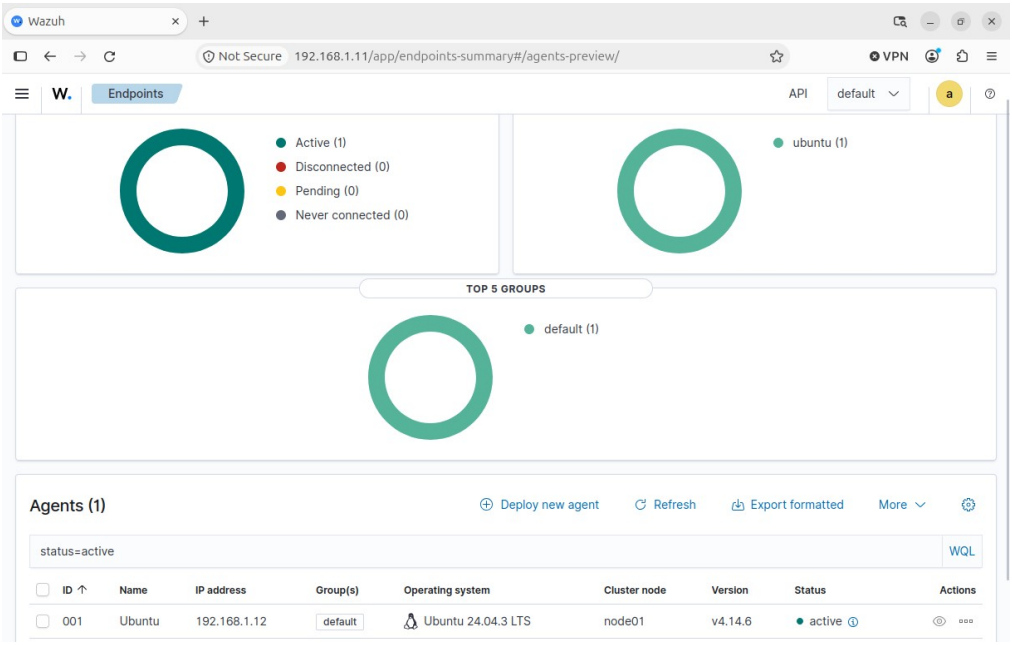
pfSense / VirtualBox	Virtual network gateway and virtualisation; existing topology preserved

Time note: Ubuntu displayed Africa/Lagos (UTC+1) while the Wazuh manager displayed UTC in the source report; timestamps should be interpreted using their displayed time zones.

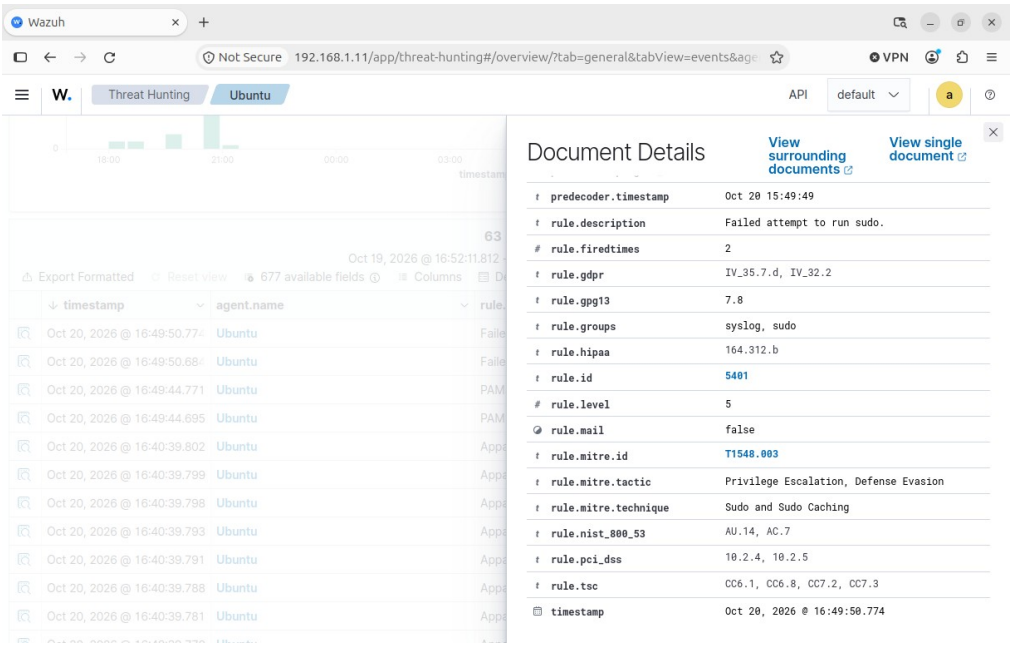
Task 5 — Endpoint monitoring and authentication telemetry

We verified Wazuh agent connectivity and service health, enabled NTP and checked system clock synchronisation. A controlled failed sudo authentication test used sudo -k followed by sudo ls /root with an incorrect password. Wazuh recorded “Failed attempt to run sudo” (rule 5401, level 5). This proves collection of authentication telemetry, not successful privilege escalation.

Technical interpretation: sudo -k invalidates the cached sudo authentication timestamp; sudo ls /root attempts to list the root user’s home directory with elevated permissions.



Task 5 evidence — Wazuh agent connectivity.



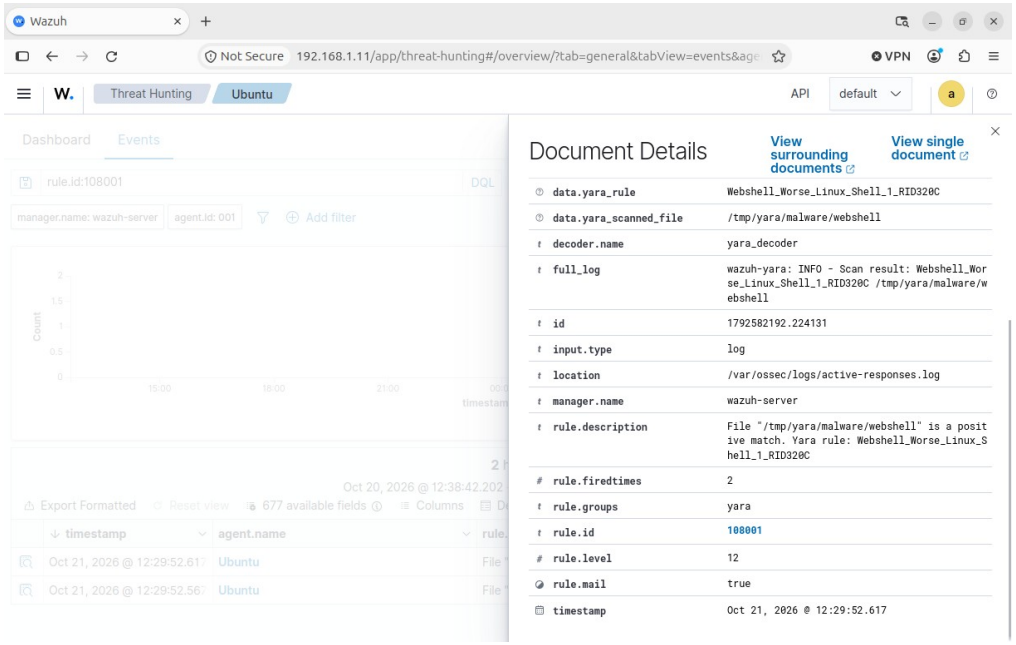
Task 5 evidence — failed sudo authentication event.

Task 6 — YARA malware-pattern detection

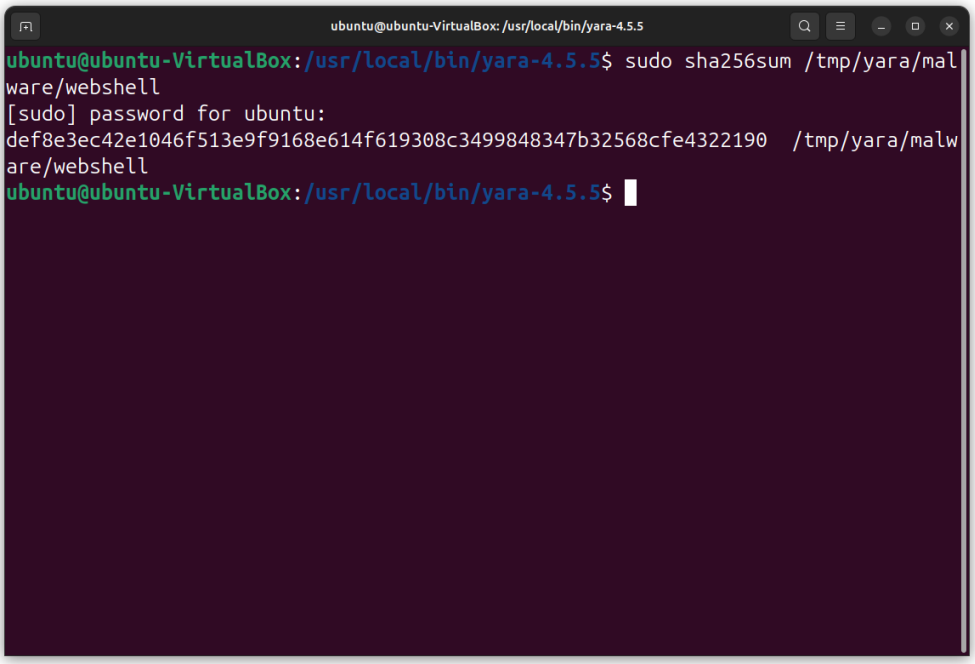
We installed and verified yara and yarac version 4.5.5. Wazuh File Integrity Monitoring observed /tmp/yara/malware, with custom file-change rules 100300 and 100301. The configured active-response script at /var/ossec/active-response/bin/yara.sh initiated scans. A simulated webshell produced a positive YARA alert (rule 108001, level 12), matching Webshell_Worse_Linux_Shell_1_RID320C.

SHA-256 for /tmp/yara/malware/webshell:
def8e3ec42e1046f513e9f9168e614f619308c3499848347b32568cfe4322190

Interpretation: FIM reports a file change; YARA examines file content against defined signatures. The evidence demonstrates scanning and detection, not automatic deletion or quarantine.



Task 6 evidence — YARA signature and detection details.



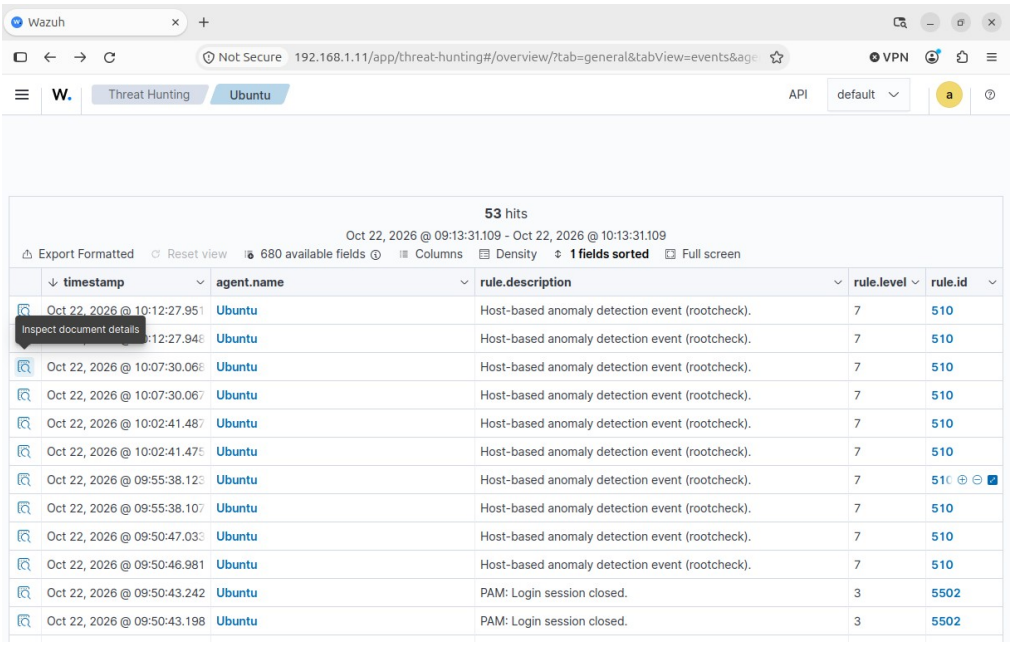
Task 6 evidence — SHA-256 collection for the simulated webshell.

Task 7 — Rootcheck suspicious binary

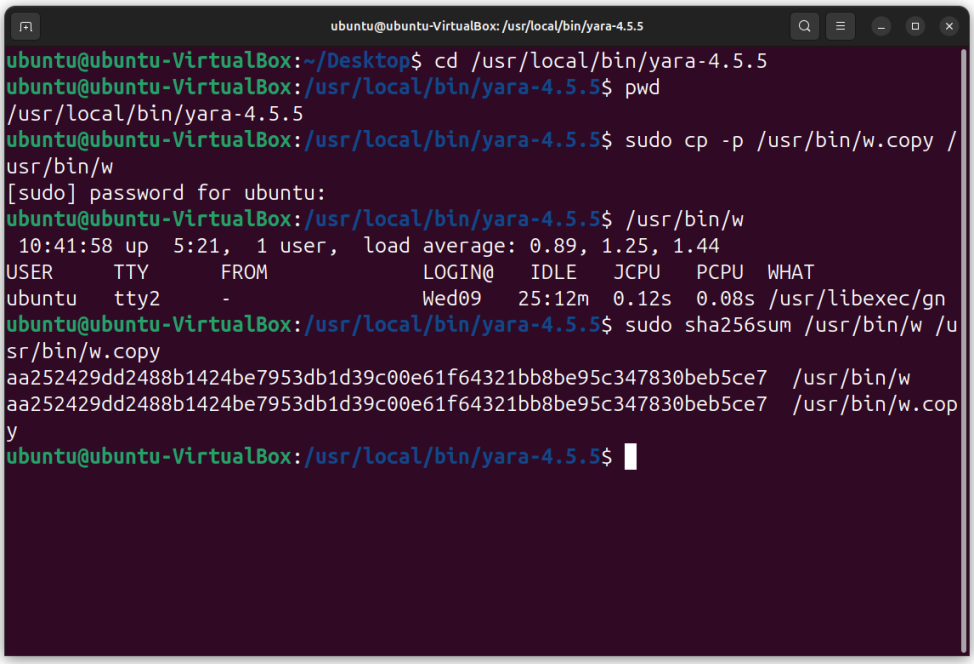
We temporarily configured Rootcheck scans every 120 seconds, preserved /usr/bin/w as /usr/bin/w.copy and replaced the system utility with a controlled test script. Wazuh produced a host-based anomaly alert (rule 510, level 7), describing a trojaned /bin/w utility. The original utility was restored manually and its SHA-256 digest matched the saved backup. The scan interval was returned to 43,200 seconds.

Verified SHA-256: aa252429dd2488b1424be7953db1d39c00e61f64321bb8be95c347830eb5ce7

Limitation: The source evidence does not independently establish a subsequent clean Rootcheck scan.



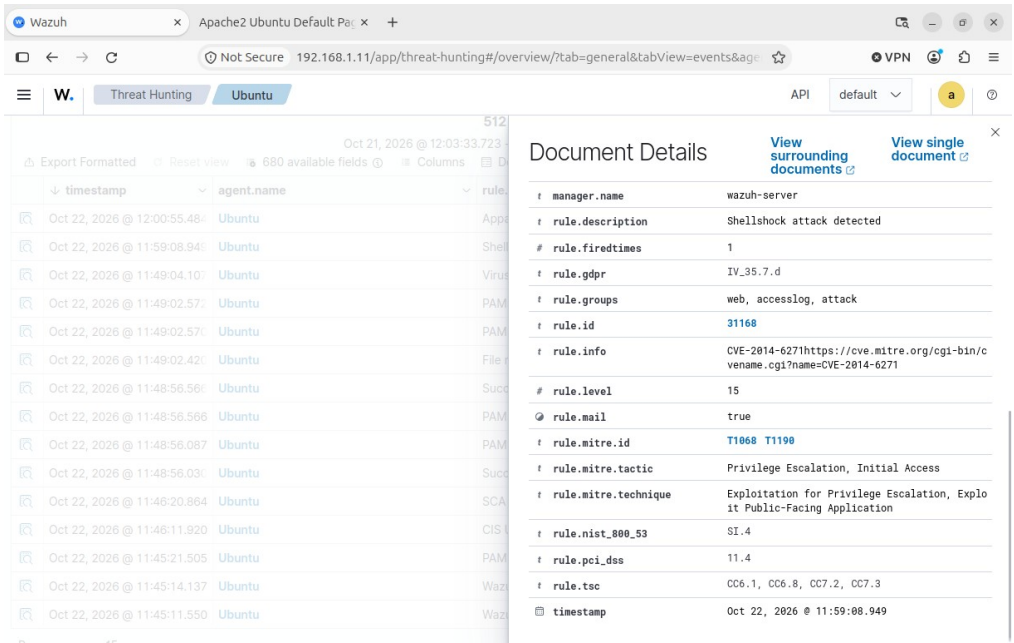
Task 7 evidence — Rootcheck alert.



Task 7 evidence — matching SHA-256 values after restoration.

Task 8 — Shellshock-style web attack detection

We monitored /var/log/apache2/access.log and sent a crafted HTTP request from Kali to the Ubuntu Apache server. Wazuh classified the request as a Shellshock attack pattern (rule 31168, level 15). A local Bash test displayed “Bash test completed” rather than “VULNERABLE.” HTTP 200 and an alert do not establish unintended command execution or successful exploitation.



Task 8 evidence — Wazuh Shellshock classification.

Consolidated technical outcomes

Task	Observed outcome	Limit
5	Agent active; sudo alert 5401 / level 5	Not proof of privilege escalation
6	YARA webshell match; rule 108001 / level 12	No confirmed auto-quarantine
7	Rootcheck alert 510 / level 7; hash-matched restoration	No verified clean rescan
8	Shellshock-style request; rule 31168 / level 15	Exploit success not established

Skills demonstrated and professional relevance

- Email threat analysis: sender identity mismatch, link inspection, attachment triage and social-engineering assessment.
- SOC monitoring: endpoint enrollment, log ingestion, alert triage and interpretation of Wazuh rule IDs.
- Detection engineering: FIM configuration, YARA integration, Rootcheck and Apache access-log monitoring.
- Incident response: evidence preservation, manual restoration, hash-based verification and defensible classification.
- Reporting: distinguishing direct observations from unverified claims and identifying evidence gaps.

Recommendations and next steps

- Use trusted original .eml files and mail-gateway logs for live phishing investigations; independently validate authentication and reputation.
- For YARA, test and document quarantine or removal only under explicit controlled procedures.
- Capture a verified clean Rootcheck scan after restoration.
- For web exploitation tests, collect additional endpoint evidence before concluding commands executed.
- Publish a redacted portfolio PDF and a GitHub README that links to evidence, methodology and clear limitations.

Conclusion

The combined project demonstrates practical entry-level SOC competencies across phishing analysis and endpoint/web threat detection. Its strongest evidence includes Wazuh alerts, signature matches, file hashes and restoration checks. The portfolio intentionally avoids overstating impact: the phishing cases are synthetic, and the web exercise confirms detection of an attempt rather than successful exploitation.

Source documents and evidence provenance

Part A: Cybersecurity_Capstone_Part_A_Phishing_Investigation(1).pdf (six pages); four instructor-provided synthetic email samples.

Part B: Cybersecurity_Capstone_Part_B_Compatible(1).pdf (eleven pages); Tasks 5–8 technical report and supplied screenshots. The report describes 24 archived screenshots, with selected evidence reproduced here.

No external threat-intelligence lookups were used in compiling this portfolio.